

INTRODUCTION

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This volume collects almost all the papers presented in the final colloquium *Image in Science and Art* that took place in Lisbon, between 17th and 19th of February, 2011, at the Calouste Gulbenkian Foundation. This colloquium was put forward by a research project funded by the Portuguese Foundation for Science and Technology – «Image in Science and Art» – which I had the pleasure to coordinate from 2007 to 2011 at the Center for Philosophy of Science of the University of Lisbon.

The project was extremely ambitious. With a large interdisciplinary team coming from Philosophy, Sciences and Arts¹, the project has elected image as its central object of research aiming to question its philosophical, scientific and artistic nature. The project was organized in twelve diversified tasks, all trying to get a deeper understanding of the status and functions of images in Science and Art. Six of those tasks were of theoretical nature endeavoring: 1) to question the concept of image and to contribute for the deepening of its theoretical approach (*For a Philosophy of Image*); 2) to analyze the nature of image in three epistemological and semiotic programs, namely those of Leibniz, Frege and Peirce (*Image and Diagrammatic Thinking*); 3) to discuss the transformations of the status and the functions of image in the passage from a word civilization to an image civilization and to evaluate the actual role of image (*From Word Civilization to Image Civilization*); 4) to question the nature of image in dialogue with some of major philosophical developments in the 20th

¹ The project's team was composed by 16 investigators, 28 national and 3 international collaborators, 9 international consultants, 3 research grant holders and 5 initiation grant holders.

century, namely Bachelard and Deleuze (*Philosophical Studies on Image between Science and Art*); 5) to face the four paradigmatic works of Leonardo, Vesalius, Goethe and Darwin in what concerns the hybrid status they attribute to image as knowledge procedures endowed of aesthetic value versus artistic objects cognitively oriented (*Historical Studies on Image between Art and Science*); 6) to discuss the capacity of recent research in Neurosciences which seeks to explain art and aesthetics by exploring the neurological basis of production and fruition of artworks (*Neuroaesthetics*).

The other six tasks were of more practical character, involving the development of empirical, comparative research. They intended 7) to investigate the role played by images in science, both in its cognitive dimension (to distinguish several images science makes use and to understand the knowledge procedures involved) and in its communicative dimension (by the exhaustive summon up of their place and role in the celebrated journal *Nature*. (*The Art of Scientific Image*); 8) to carry out a systematic study of scientific illustration in Portugal from the 16th century to the actuality (*Drawing Science*); 9) to identify, to catalogue and to study the unexplored collection of anatomical drawings belonging to the Museum of Medicine of Lisbon University (*Anatomical Drawing in the Collection of the Museum of Medicine*); 10) to confront the scientific study of human anatomy with the artistic drawing of human body through the study of the nude drawing's collection belonging to the Fine Arts School of the University of Lisbon (*Drawing the Nude*); 11) to built an online repository of scientific and artistic images, opened to artistic, scientific and philosophical communities, that is, a dynamic archive intended to select, to compare and to contrast a set of scientific and artistic images in order to look for their differences and inner affinities (*Lisbon Center for Image in Science and Art*).

The last task – *Crossings* – intended the organization of a full program of events to provide a large dissemination of the project's work and results. In this context, a series of four large audience conferences was organized in Lisbon at the Calouste Gulbenkian Foundation: *Taking it on Trust' in Images of Nature*, by Martin Kemp (November 17th, 2010); *The Problem of a Picture of an Atom*, by Chris Toumey (December 15th, 2010); *Visiting Time: the Renegotiation of Time through Time-based Art*, Boris Groys

(January 19th, 2011); *Functional Images of the Brain: Beauty, Bounty, and Beyond*, by Judy Illes (February 2nd, 2011). A final exhibition – *BodyIMAGE. Body Representations in Science and Art* – was put forward at the Knowledge Pavillion – Ciência Viva. Open to the public from 1st February up to 30th April 2011, the exhibition included images of the human body produced by Science and Art over the last 150 years in Portugal. It was organized in three main moments following the technical and technological decisive transformations in the image technology, trying to evaluate their impact in the production of body images in Science and Art². Finally, an international colloquium entitled *Image in Science and Art* (February 17th-19th, 2011) was organized at the Calouste Gulbenkian Foundation, of which we publish here the proceedings. Other achievements may also be assessed by the project's productivity both in terms of the organization of scientific events from 2007 to 2011,³ of the edition of a series of twelve volumes untitled “Image in Science and Art”⁴ and of several other publications⁵.

² The first moment was constituted by drawings; the second by photography and x-ray images, the third by digital imaging. At the first moment, the anatomist and the artist had only at their disposal that virtuous instrument of observation, description and representation which is the carbon and the pencil. At the second moment, the photographic camera and the x-ray apparatus were able to catch analogical images which have an enormous effect over medical knowledge and pictorial representation of the body. At the third, the image of the body is not captured but created, computationally produced on the basis of numerical data collected by diverse new technologies. Scientists and artists will be able take the consequences of those transformations.

³ Please see the project's webpage: <http://ica.fc.ul.pt>

⁴ The serie is being edited by Fim de Século and includes: 1) João Araújo (2010), *As Cidades, os Castelos e a Onda – Imagens, diagramas e metáforas entre Calvino, Escher e Bohr*; 2) Olga Pombo, Silvia Di Marco and Marco Pina (Org.) (2010), *Neuroaesthetics. Can Science explain Art?*; 3) Olga Pombo and Silvia Di Marco (Org.) (2010), *As Imagens com que a Ciência se Faz*; 4) Rodrigo Vilhena (2011), *Solares. Sistema Beta Pictoris*; 5) Alberto Faria (2011), *A Coleção de Desenho Antigo da Faculdade de Belas-Artes de Lisboa (1830-1935). Tradição, Gosto e Formação*; 6) Cristina Azevedo Tavares (Org.) (2012), *O Desenho do Corpo na Arte e na Ciência*; 7) Olga Pombo and Marco Pina (Org.) (2012), *Em torno de Darwin*; 8) Nuno Melim (2013), *Visível & Enunciável – Uma derrota por Foucault*; 9) Olga Pombo and António Guerreiro (Org.) (2013), *Da Civilização da Palavra à Civilização da Imagem*; 10) Olga Pombo (Org.) (2016), *Image in Science and Art*; 11) Olga Pombo (Org.), *Lugares e Modelos* (in press); 12) Olga Pombo, Catarina Nabais e Sara Fuentes (Org.), *Body-Image. Crossings, Catálogo de uma Exposição* (in press).

⁵ Behind the publication of many articles and chapters (cf. <http://ica.fc.ul.pt>), the team members of the project published the following books: 1) Ana Gaspar (Ed.) (2008), *Bachelard: Epistemologia, Ciência e Poética*, 6.º Caderno de Filosofia das Ciências, Lisboa: CFCUL; 2) Nuno Carvalho (2009), *A imagem-sensação: Deleuze e a pintura*, Coleção Thesis n.º 2, Lisboa: CFCUL; 3) Ana Gaspar (2010), *Entre o Conceito e a Imagem. O lugar da Psicanálise na obra de Gaston Bachelard*, Lisboa: CFCUL; 4) Olga Pombo and Alexander Gerner (Ed.) (2010), *Studies in Diagrammatology and Diagram Praxis*, London. College Publications; 5) Catarina Nabais (2013), *Gilles Deleuze: Philosophie et Littérature*, Paris: Harmattan; 6) Pedro Salgado, *Cinco Séculos de Ilustração Científica em Portugal*, Lisboa: Assírio & Alvim (in press).

The colloquium was a remarkable event of high scientific level. It was structured in six sessions, each of which dedicated to a major topic. The colloquium begun by an opening ceremony in which several personalities took the word, namely the President of the University of Lisbon, at the time, who made a brief presentation which we are honored to include in these proceedings. The first session, on the morning of February 17th – *Philosophy of Image* – was chaired by me, Olga Pombo. It counted with two opening conferences: *Pause of Participation. On the Function of Artificial Presence*, by Lambert Wiesing, and *Bild und Bildung: Seeing and knowing from the viewpoint of a neuroscientist*, by Ernest Pöppel. These two conferences were followed by a debate between Lambert Wiesing and Ernest Pöppel moderated by Alexander Gerner. The second session, in the afternoon of the same day, was dedicated to the specific case of *Nanoimages*. Coordinated by Rasmus T. Slaattelid, this session included four conferences, three of scientific nature: *Imaging nanostructures and nanodevices*, by Paulo Treitas; *Different ways of looking at things*, by Bodil Holst; *A brief phylogenesis and ontogenesis of nanoimages*, by Alice Benessia. The fourth conference, which we are very much pleased to publish now, was offered by the nano-artist, Chris Robinson, under the title *The artist as mediator in seeing and understanding nanotechnology*. The third session which took place at the morning of the second day, February 18th, was focused on the so-called *Civilization of Image*. It included two conferences: *Silent Story-Telling: Typologies of Iconic Narratives*, by Andrea Pinotti, and *Image in theatre – The case of 'O Bando'*, by Nuno Nabais. In that session also took place a splendid round-table dedicated to Pedro Costa's movies. Chaired by António Guerreiro, the round-table included a conference by Jacques Rancière followed by a debate with Jacques Rancière and Pedro Costa. All the materials have been recorded and are here published. The fourth session, at the afternoon of February 18th – *Visualization of Complexity* – was chaired by Manuel Lima who kindly accepted to picture and to organize all the session. Manuel Lima made a presentation on *The Emergence of Visualization* followed by the talks of the three invited researchers and creative practitioners on visualization studies: Jeff Thorp, Miguel Cardoso and Henry Markram who presented *Time-Based Visualization of Complex Systems, Interaction and Complexity in Information Visualization*, and *Visualization of complexity: expressiveness and accuracy*, respectively. The fifth session – *Science and*

Art – took place at the morning of the last day, February 19th, 2011. It was thoughtfully coordinated by Alexandre Quintanilha. The session was organized with two duos, each of which with a scientist and an artist. The first duo was carried out by the biologist Rui Malhó and the photographer and ceramist Rob Kessler, both dealing with pollen, respectively *Science and art: are creativity and rigor in opposite corners?* and *Science and sensibility: an artist approach to micro imaging*. The second duo was achieved by the neuroscientist Daniela Seixas and the bio-artist Marta de Menezes, both on brain imaging. Their talks were respectively entitled *Neuroimaging the brain* and *The art of ideas and the ideas of art*. During the colloquium a very original presentation of Rob Kessler's ceramics ornamented with pollen images was offered in the atrium of the Calouste Gulbenkian Foundation. The session also included a closing conference by Jorge Calado called *Science-art continuum*. The sixth and last session – *Scientific Illustration* – took place at the afternoon of the last day. It was chaired by the biologist and scientific illustrator Pedro Salgado and totally dedicated to the Portuguese school of scientific illustration. It included a video presentation on the history of Portuguese illustration from the 16th century until nowadays. The contents of the video – *Scientific Illustration from Garcia da Horta to Risk's Group* – was afterwards commented by Pedro Salgado, Diana Marques, Nuno Farinha and Ana Teresa Bígio. We are very happy for having the possibility of publishing all the talks given: *Scientific illustration overview – Subject matters and applications*, by Diana Marques; *Pioneering modern scientific illustration – Five centuries of drawings of Portuguese Natural History*, by Nuno Farinha; *Scientific illustration in scientific education in Portugal*, by Ana Teresa Bígio; and *Field sketching and scientific illustration: drawing to understand nature and drawing to explain*, by Pedro Salgado. The colloquium ended with a final debate and a closing ceremony.

Overall, all along the three days of the colloquium, 22 excellent conferences and one unforgettable round-table were presented. And, if we regret that some of the presentations were not sent to us for publication (namely those by Paulo Freitas, Bodil Holst, Alice Benessia, Jeff Thorp, Miguel Cardoso, Henry Markram, Rui Malhó and Daniela Seixas), we are very happy for having the possibility of recovering in the proceedings most of those conferences given in the colloquium. A very special word is due in memory of Hermínio Martins, an emeritus fellow authority on sociology.

In fact, even he could not managed to participate in the colloquium, Hermínio Martins kindly sent me his excellent text for publication. Something which is now done, after his unfortunate disappearing.

I profit the opportunity to gratefully thank all those who contributed to the success of the colloquium. In doing so, I am pleased to thank João Caraça, director of the science department of Calouste Gulbenkian Foundation who received the idea of the colloquium with great enthusiasm and gave it a fully generous support, the President of the University of Lisbon, António Nóvoa who kindly accepted to participate in the colloquium and wanted to contribute with a foreword. I also thank all those who most competently accepted to chair the several sessions, Alexandre Quintanilha, António Guerreiro, Manuel Lima, Nuno Nabais, Pedro Salgado, Rasmus T. Slaattelid. My deep gratitude to all our invited speakers for their availability in sharing with us their expertise, Lambert Wiesing, Ernst Pöppel, Paulo Freitas, Bodil Holst, Alice Benissia, Chris Robinson, Andrea Pinotti, Nuno Nabais, Pedro Costa, Jacques Rancière, Manuel Lima, Jer Thorp, Miguel Cardoso, Pedro Miguel Cruz, Rui Malhó, Rob Kessler, Daniela Seixas, Marta de Menezes, Jorge Calado, Diana Marques, Nuno Farinha, Ana Teresa Bígio and Pedro Salgado.

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